

The Collectible Market Bubble and Burst: Evidence from the Pre- and Post-COVID Stimulus Cycle, 2020–2025

Arthur Mota

Independent Research

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Abstract

We study the association between COVID-era household liquidity growth and price dynamics in collectible asset markets: pre-2000 Magic: The Gathering (MTG) Reserved List cards and pre-2000 classic automobiles. MTG cards appreciated 165% (cumulative) during 2020 Q2–2021 Q4, classic cars 32%, versus 57% for the S&P 500. We document this association through four strategies: discrete regime breaks, within-regime demeaned household liquidity from the Fed’s Z.1 Financial Accounts, Granger causality in an expanded VAR, and cross-sectional variation in transaction frictions. The stimulus regime dummy is associated with 12.3 pp/quarter of excess MTG returns ($t = 3.35$, Newey-West HAC), and within-regime demeaned liquidity remains significant ($t = 2.36$), ruling out spurious co-trending. MTG exhibits strong positive return persistence ($t = 4.95$) absent in equities ($t = -1.27$), consistent with speculative feedback dynamics rather than standard portfolio rebalancing. M2 growth Granger-causes MTG returns in a VAR(1) that includes fiscal transfer flows and the FFR level ($p = 0.027$). A seven-component, source-documented friction index shows suggestive evidence that higher-friction markets experienced weaker stimulus responses ($\hat{\beta} = -0.094$, 95% CI: $[-0.201, 0.014]$). For classic cars, the semiconductor chip shortage dominates: a chip-shortage dummy absorbs the stimulus dummy. Leave-one-out diagnostics confirm the MTG stimulus coefficient is stable (range 0.110–0.154). The subsequent hiking cycle reversed much of the gains. The results indicate that when fiscal authorities inject trillions into household balance sheets with consumption constrained by lockdowns, a measurable fraction flows into speculative alternative assets, with the magnitude moderated by transaction frictions.

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1 Introduction

The fiscal and monetary policy response to the COVID-19 pandemic was without precedent in U.S. economic history. Between March 2020 and March 2021, Congress enacted approximately \$5 trillion in fiscal support, including direct transfers, enhanced unemployment benefits, and Paycheck Protection Program loans, while the Federal Reserve slashed the federal funds rate to the zero lower bound and expanded its balance sheet by about \$3.5 trillion (\$4.7 trillion by early 2022) through asset purchases. M2 money supply surged from \$15.3 trillion in 2019 Q4 to \$21.4 trillion by 2021 Q4, a 40% increase in eight quarters.

The effects of this liquidity injection on traditional financial assets are well documented: the S&P 500 recovered its pre-pandemic highs within five months and reached all-time records by late 2020. Bitcoin, the canonical speculative asset, rose from roughly \$7,600 in 2019 Q4 to \$51,000 by 2021 Q4. What has received less systematic attention is the transmission of this liquidity shock to *illiquid collectible* markets, i.e. assets with no cash flows, no fundamental valuation anchor, and price discovery driven almost entirely by collector sentiment and perceived scarcity.

This paper fills that gap by studying two collectible asset classes: (1) pre-2000 Magic: The Gathering (MTG) cards on the Reserved List, a contractually supply-constrained set of cards that Wizards of the Coast has committed never to reprint, and (2) pre-2000 classic automobiles, specifically Porsche 911 variants (SC, Targa, Carrera 3.2, 964, 993) and Ford Mustang models (1967–1979 Fastback, Mach 1, and Mustang II). Both markets are characterized by fixed supply, heterogeneous quality, information asymmetry, and thin trading, features that make them particularly susceptible to sentiment-driven price dislocations.

Our central hypothesis is that excess liquidity and direct fiscal transfers generated speculative demand in collectible markets, as retail participants with unusually high liquid savings and reduced consumption opportunities redirected capital toward alternative stores of value. We distinguish this speculative demand channel from the traditional portfolio balance channel by testing for return persistence (momentum) in collectible prices: speculative feedback implies that lagged returns predict current returns as participants chase recent appreciation, whereas a pure wealth effect does not generate return persistence. We find that MTG exhibits strong positive return persistence ($t = 4.95$) that is absent in equities ($t = -1.27$), consistent with speculative demand dynamics. We use the term “FOMO” informally to describe this speculative feedback mechanism but note that the empirical content of our claim rests on the momentum test and the regime identification, not on a formal model of belief contagion.

We organize the analysis around three monetary policy regimes: (1) the *pre-COVID baseline* from 2015 Q1 to 2020 Q1; (2) the *stimulus period* from 2020 Q2 to 2021 Q4; and (3) the *hiking cycle* from 2022 Q2 to 2023 Q3, during which the Federal Reserve raised the federal funds rate from near zero to 5.33%.

1.1 Related Literature

The relationship between monetary policy and asset prices is a cornerstone of macroeconomic research. [Bernanke and Kuttner \(2005\)](#) demonstrate that unanticipated easing raises equity prices, mainly through lower expected excess returns; [Rigobon and Sack \(2004\)](#) identify the structural response of equity and bond prices to monetary policy shocks using heteroskedasticity-based identification. More recently, [Greenwood et al. \(2022\)](#) document the role of credit supply in driving asset price cycles.

The collectibles literature is thinner but growing. [Mei and Moses \(2002\)](#) study the long-run returns to art and find that art underperforms equities on a risk-adjusted basis but exhibits low correlation with financial markets. [Korteweg et al. \(2016\)](#) develop methods for estimating returns in markets with infrequent and selection-biased transactions, directly applicable to

collectibles. [Renneboog and Spaenjers \(2013\)](#) provide evidence that art prices co-move with equity markets over long horizons but exhibit substantial idiosyncratic risk.

On the behavioral side, [Baker and Wurgler \(2007\)](#) document that investor sentiment predicts cross-sectional returns in equities; the FOMO phenomenon in alternative assets during COVID has been noted in journalistic accounts but lacks formal econometric treatment. The present paper bridges these literatures by bringing standard macro-finance empirical tools to bear on collectible price data spanning the full COVID policy cycle.

The present paper is macro in orientation: we treat collectible price indices as dependent variables and monetary/fiscal policy measures as the primary independent variables of interest.

1.2 Theoretical Framework

We distinguish three channels through which a positive liquidity shock can raise collectible prices, each with distinct empirical predictions.

Channel 1: Portfolio rebalancing (wealth effect). A standard portfolio balance channel predicts that an increase in household liquid assets reduces the marginal utility of cash holdings, causing reallocation toward riskier assets. This channel predicts: (a) a level shift in collectible prices proportional to the liquidity shock, (b) no return persistence beyond what the liquidity variable explains, and (c) a symmetric reversal when liquidity contracts. If this channel alone were operative, the regime dummy should be insignificant once household liquidity growth is included as a continuous regressor, and lagged returns should not predict current returns.

Channel 2: Speculative feedback (return-chasing). Building on [DeLong et al. \(1990\)](#) and [Baker and Wurgler \(2006\)](#), speculative demand arises when rising prices attract new participants who observe recent appreciation and enter the market to capture further gains, bidding prices above what fundamentals (including liquidity) justify. This channel predicts: (a) a regime dummy that remains significant even after controlling for liquidity, capturing the self-reinforcing component, (b) positive return persistence (lagged returns predict current returns) as participants chase recent performance, and (c) potentially asymmetric dynamics if selling frictions or loss aversion impede exit. The empirical signature that separates Channel 2 from Channel 1 is the momentum test: significant positive return autocorrelation in collectibles that is absent in efficiently priced assets.

Channel 3: Transaction friction moderation. Both channels 1 and 2 predict that asset classes with lower transaction frictions should exhibit stronger price responses to the same aggregate liquidity shock, because frictions impede the rate at which new capital can be deployed. The friction interaction tests this prediction cross-sectionally.

Our identification strategy combines time-series regime identification (testing for a discrete shift in returns), within-regime liquidity variation (testing for a continuous relationship), momentum diagnostics (separating speculative feedback from portfolio rebalancing), and cross-sectional friction variation. We do not claim that any single test achieves clean causal identification in isolation. The COVID stimulus window coincides with zero rates, QE, fiscal transfers, lockdowns, reduced services consumption, and changed time allocation. Our approach is to use multiple strategies that are vulnerable to different confounders, such that the joint evidence is more informative than any individual test.

2 Data

2.1 Collectible Price Indices

MTG Reserved List Index. We construct a quarterly price index for a basket of pre-2000 Reserved List cards, including dual lands (Underground Sea, Volcanic Island, Tropical Island, Bayou, Savannah, Tundra, Scrubland, Badlands, Plateau, Taiga), Power Nine components (Black Lotus via Unlimited Edition, Mox Pearl, Mox Sapphire, Mox Jet, Mox Ruby, Mox

Emerald, Ancestral Recall, Time Walk, Timetwister), and other Reserved List staples (The Tabernacle at Pendrell Vale, Moat, Gaea’s Cradle, Lion’s Eye Diamond). Prices are sourced from TCGPlayer market prices and MTGGoldfish price histories, with quarterly observations taken at the mid-quarter date. The index is normalized to 100 at 2015 Q1. Reserved List cards are supply-constrained by design: Wizards of the Coast’s 1996 Reserved List policy guarantees these cards will never be reprinted, making them the closest analog to a fixed-supply commodity in the collectibles space.

Classic Car Indices. Quarterly driver-quality values for air-cooled Porsche 911 variants (1978–1998) (SC Coupe, SC Targa, Carrera 3.2 Coupe, 964 Carrera 2, 993 Carrera 2) and pre-1979 Ford Mustang variants (1967–68 Fastback V8, 1969–70 Mach 1, 1969–70 Fastback V8, 1971–73 Mach 1, Mustang II V8) are interpolated from annual Hagerty and Classic.com anchor points, supplemented with Bring a Trailer and Mecum Auctions transaction data. We construct equal-weighted sub-indices for each marque and a composite index, all normalized to 100 at 2015 Q1.

2.2 Macro and Financial Variables

All macro data are at quarterly frequency, 2015 Q1–2025 Q4:

- **Federal funds rate (FFR):** Effective federal funds rate, quarterly average. Source: FRED (FEDFUNDS).
- **M2 money supply:** Broad money stock in trillions of USD, end-of-quarter. Source: FRED (M2SL). Used in the VAR and excess-return regressions and as a comparison to the Z.1 measure.
- **Household liquid assets (Z.1):** Sum of household checkable deposits and currency (BOGZ1FL153020005Q), time and savings deposits (BOGZ1FL153030005Q), and money market fund shares (BOGZ1FL153034005Q), in billions of USD, end-of-quarter. Source: Federal Reserve Z.1 Financial Accounts of the United States, Table B.101.
- **Household checkable deposits (Z.1):** Checkable deposits and currency held by households and nonprofit organizations (BOGZ1FL153020005Q), in billions of USD. Source: Federal Reserve Z.1 Financial Accounts.
- **S&P 500:** Quarterly average level. Source: S&P Dow Jones Indices.
- **Bitcoin (BTC):** Quarterly average price in USD. Source: CoinMarketCap.
- **VIX:** CBOE Volatility Index, quarterly average. Source: FRED (VIXCLS).
- **Consumer Sentiment:** University of Michigan Index of Consumer Sentiment, quarterly average. Source: University of Michigan / FRED (UMCSENT).
- **Fiscal stimulus:** Flow of federal COVID-related transfers per quarter (billions USD), cumulated for stock measure. Sources: CBO, Treasury Department.
- **Manheim Used Vehicle Value Index:** Wholesale used vehicle price index, quarterly average. Base: January 1995 = 100. Source: Manheim / Cox Automotive.
- **CPI Used Cars and Trucks:** BLS consumer price index for used cars and trucks (CUSR0000SETA02), quarterly average. Source: Bureau of Labor Statistics.

Household liquidity vs. M2 money supply. We use household-level balance sheet data from the Federal Reserve’s Z.1 Financial Accounts (Flow of Funds) rather than M2 money supply as our preferred liquidity measure. The rationale and empirical dominance of Z.1 over M2 are discussed in Section 3.1. Household checkable deposits surged from \$1.17 trillion in 2019 Q4 to \$3.80 trillion by 2021 Q4 (a 225% increase), while total household liquid assets rose from \$14.1 trillion to \$19.9 trillion.

Supply-side control: the semiconductor shortage. A critical confound for classic car price analysis during 2020–2022 is the global semiconductor chip shortage, which severely constrained new vehicle production. With new car inventory at historic lows, demand spilled into the used vehicle market, driving the Manheim Used Vehicle Value Index from 132 in 2020 Q1 to a peak of 236 in 2022 Q1, a 79% increase. While classic pre-2000 automobiles do not directly compete with modern used vehicles, they exist within the same broader automotive ecosystem: rising used car prices increase the opportunity cost of selling a classic, elevate the general “car as asset” narrative, and attract marginal buyers who perceive all vehicles as appreciating assets. We therefore include the Manheim index return as a supply-side control in our car regressions to separate the chip-shortage channel from the household liquidity channel.

2.3 Derived Variables

We compute log returns $r_t = \ln(P_t/P_{t-1})$ for all price indices, quarterly M2 growth rates, and the first difference of the federal funds rate. All regression specifications use log returns as the dependent variable. Cumulative percentage returns reported in narrative text (e.g., “MTG appreciated 165%”) are simple cumulative returns $(P_T/P_0 - 1) \times 100$ and are provided for economic interpretation only; they do not enter any regression. The conversion between conventions is monotonic and near-linear over small quarterly intervals but diverges for large cumulative moves; we use the log specification throughout for econometric consistency. Regime dummy variables are defined as follows: $COVID_STIM_t = 1$ for 2020 Q2 through 2021 Q4; $HIKE_t = 1$ for 2022 Q2 through 2023 Q3.

2.4 Descriptive Statistics

Table 1 reports summary statistics for the full sample.

Table 1: Summary Statistics, 2015 Q1–2025 Q4 (44 Quarterly Observations)

	Mean	Std. Dev.	Min	Max	Median
MTG Index	304.1	156.6	100.0	640.0	260.0
Porsche Index	200.4	71.1	100.0	296.9	207.7
Mustang Index	136.7	21.0	100.0	172.1	140.1
Car Composite	168.6	45.6	100.0	234.5	177.1
S&P 500	3,557	1,226	1,951	5,890	3,240
Bitcoin (USD)	25,314	27,523	240	92,000	11,300
Fed Funds (%)	2.02	1.95	0.06	5.33	1.34
M2 (\$T)	17.37	3.77	11.85	21.90	18.38
VIX	18.5	6.0	9.5	33.1	16.0
Consumer Sent.	80.3	15.5	52.0	99.2	82.1

Notes: MTG, Porsche, Mustang, and Car Composite indices normalized to 100 at 2015 Q1. S&P 500 and Bitcoin in levels. Fed funds rate and VIX are quarterly averages.

Table 2 reports return correlations.

Table 2: Quarterly Return Correlations

	MTG	Cars	S&P 500	Bitcoin
MTG	1.000	0.726	0.534	0.498
Cars		1.000	0.255	0.279
S&P 500			1.000	0.491
Bitcoin				1.000

Several features are notable. MTG and classic car returns are strongly correlated (0.73), suggesting a common collectibles factor. MTG returns are more correlated with equities (0.53) and Bitcoin (0.50) than car returns are, consistent with a younger, more financially active collector base in the MTG market. The correlation between MTG and Bitcoin is suggestive of a shared speculative demand driver.

3 Methodology

3.1 Identification Strategy

Estimating the association between stimulus and collectible prices faces a standard macro-finance challenge: household liquidity and asset prices both trended upward over the sample, raising the concern that any regression of returns on liquidity growth captures co-movement of trends rather than a genuine economic channel. We address this through four complementary identification strategies.

First, we exploit the discrete regime break at 2020 Q2. The COVID stimulus was an exogenous policy shock driven by pandemic response, not by collectible market conditions. The regime dummy $STIM_t$ captures the average excess return during the stimulus window after controlling for continuous macro variables. Because it is a binary shift, it is immune to the trending-regressor problem.

Second, we demean continuous regressors within each monetary policy regime. For any variable x_t , we compute $\tilde{x}_t = x_t - \bar{x}_{r(t)}$ where $r(t) \in \{\text{pre-COVID, stimulus, hike, post-hike}\}$. This strips out between-regime level shifts and isolates within-regime variation: if the demeaned liquidity coefficient remains significant, it means that quarters with above-average liquidity growth *within a given regime* predict higher collectible returns, a relationship that cannot be driven by trending.

Third, we exploit cross-sectional variation in transaction frictions across five asset classes (Section 3.4). Because frictions are time-invariant structural features of each market, the friction $\times$ stimulus interaction identifies the differential impact of the same aggregate liquidity shock across markets with different ease of access, providing a clean cross-sectional source of identification.

Fourth, we control directly for the “boredom channel” by including quarterly Google Trends search interest growth for “Magic the Gathering” and “classic cars” as regressors. If the pandemic-driven increase in screen time and hobby engagement, rather than liquidity, drove collectible prices, the Google Trends proxy should be significant and the liquidity coefficient should attenuate. We show that the Google Trends controls are insignificant ($t = -0.84$ for MTG, $t = -1.47$ for cars) while the liquidity coefficient survives and strengthens.

Z.1 vs. M2 as the liquidity measure. M2 money supply is the standard aggregate monetary measure, but it is a poor proxy for the specific channel at work here: retail purchasing power available for speculative deployment. M2 includes components (business and other non-household deposits) with no direct bearing on household spending decisions, and its predictive power for inflation has been unstable since the 1990s. The Z.1 Financial Accounts measure

directly captures the liquid assets sitting in household checking accounts, savings accounts, and money market funds, where fiscal transfers landed almost dollar-for-dollar. Empirically, Z.1 dominates M2 in all specifications: lower AIC/BIC, higher adjusted R^2 , and stronger t -statistics. When both are included in the same regression, M2 flips sign and becomes insignificant.

All standard errors in the preferred specifications are Newey-West HAC with three lags, correcting for both heteroskedasticity and residual autocorrelation. We report Durbin-Watson statistics, Cook’s distance, and leave-one-out sensitivity throughout. We also report influence diagnostics and parallel trends tests for the DiD specification.

3.2 OLS with Regime Controls

Our baseline specification regresses collectible returns on macro factors and regime dummies:

$$r_t^{\text{coll}} = \alpha + \beta_1 \Delta \ell_t + \beta_2 \Delta i_t + \beta_3 r_t^{SP} + \beta_4 \text{VIX}_t + \gamma_1 \text{STIM}_t + \gamma_2 \text{HIKE}_t + \varepsilon_t \quad (1)$$

where r_t^{coll} is the quarterly log return on the collectible index (MTG or cars), $\Delta \ell_t$ is the quarterly growth in household liquid assets from the Z.1 Financial Accounts (our preferred specification) or alternatively M2 growth (robustness check), Δi_t is the change in the federal funds rate, r_t^{SP} is the S&P 500 return, and VIX_t is the quarterly average VIX level. We also estimate specifications using household checkable deposit growth as a narrower liquidity measure that captures the most stimulus-sensitive component of household balance sheets. All preferred standard errors are Newey-West HAC (3 lags); HC1 results are reported for comparison.

For classic car regressions, we augment the specification with a supply-side control:

$$r_t^{\text{car}} = \alpha + \beta_1 \Delta \ell_t + \beta_2 \Delta i_t + \beta_3 r_t^{SP} + \beta_4 \text{VIX}_t + \gamma_1 \text{STIM}_t + \gamma_2 \text{HIKE}_t + \phi r_t^{\text{Manheim}} + \varepsilon_t \quad (2)$$

where r_t^{Manheim} is the quarterly log return on the Manheim Used Vehicle Value Index. This controls for the global semiconductor chip shortage that disrupted new vehicle production from mid-2020 through early 2022 and drove used vehicle prices to historic highs. By absorbing the supply-side automotive price channel, any residual significance on γ_1 represents a demand-side speculative effect on classic car prices. We also estimate an alternative specification replacing the continuous Manheim control with a CHIP_SHORT_t dummy equal to one from 2020 Q3 through 2022 Q2.

The coefficient γ_1 on STIM_t captures the average quarterly excess return during the stimulus period *after controlling for* household liquidity growth, rate changes, equity market returns, volatility, and (for cars) the used vehicle supply disruption. A positive and significant γ_1 indicates that collectible prices responded to the stimulus regime above and beyond what macro fundamentals and supply conditions would predict; the residual we attribute to speculative demand and behavioral risk appetite.

3.3 Difference-in-Differences (Supporting Evidence)

As a supplementary specification, we construct a panel of collectible (MTG and car) returns and equity (S&P 500) returns stacked over time, and estimate:

$$r_{it} = \alpha + \delta_1 \text{COLL}_i + \delta_2 \text{POST}_t + \delta_3 (\text{COLL}_i \times \text{POST}_t) + u_{it} \quad (3)$$

where $\text{COLL}_i = 1$ for collectible assets and $\text{POST}_t = 1$ during the stimulus window. The coefficient δ_3 measures the excess return to collectibles relative to equities during the stimulus period.

3.4 Transaction Friction as a Treatment Moderator

A key concern is that the differential stimulus response between MTG and classic cars may reflect structural market microstructure differences rather than monetary policy transmission. Both markets have well-developed online platforms: MTG cards trade on TCGPlayer and eBay with 8–12% platform fees, while classic cars list on Bring a Trailer, Mecum Auctions, Cars & Bids, and Hemmings. The critical difference is not online access (both have it) but *post-transaction regulatory and physical friction*. MTG cards require no documentation, fit in a binder, ship for \$5, and can be bought for as little as \$1. Classic cars require DMV title transfers, state registration, insurance procurement, a pre-purchase inspection (standard practice), enclosed transport logistics (\$1,000–\$3,000 cross-country), dedicated garage storage, and cannot be divided; one must buy the entire vehicle at a \$20,000–\$250,000 price point. These frictions are *time-invariant* properties of each market; they do not change during the stimulus period. What changes is the *interaction* between fixed frictions and the time-varying liquidity shock.

We formalize this by constructing a composite friction index from seven verifiable, source-documented components: (1) platform transaction fees (% of sale price; sources: Schwab/Fidelity fee schedules, Coinbase fee page, TCGPlayer seller FAQ, Bring a Trailer terms), (2) required regulatory documents (count; state DMV title transfer requirements), (3) transport cost as % of asset value (USPS flat-rate pricing, uShip enclosed auto transport quotes), (4) storage cost as annual % of value (self-storage and climate-controlled garage market rates), (5) minimum practical ticket size (\$; minimum lot sizes on each platform), (6) settlement time in days (SEC T+2 rule, Bitcoin network confirmation, USPS Priority Mail, state DMV processing times), and (7) divisibility (binary; whether fractional purchases are possible). Each component is normalized to $[0, 1]$ across the five assets, divisibility is inverted (divisible = low friction), and the seven components are equally weighted. The resulting scores are: S&P 500 (0.9), Bitcoin (0.6), MTG (24.3), Mustang (86.3), Porsche (90.5). Table 12 in the Appendix reports the raw values and sources for each component. We then estimate a friction-moderated stimulus model:

$$r_{it} = \alpha + \beta_1 f_i + \beta_2 POST_t + \beta_3 (f_i \times POST_t) + u_{it} \quad (4)$$

where $f_i \in [0, 1]$ is the normalized friction score for asset i . The coefficient β_3 tests whether higher-friction assets experienced a weaker stimulus response, as we would expect if ease of transacting amplifies speculative capital inflows.

3.5 Vector Autoregression

We estimate two VAR specifications. The baseline VAR includes four variables (MTG returns, car returns, M2 growth, and the change in the federal funds rate). The expanded VAR adds fiscal transfer flows and the FFR level to address the concern that M2 Granger-causes MTG returns only because both are correlated with the fiscal regime; if the Granger causality from M2 survives conditioning on fiscal flows and the rate level, it is not an artifact of omitted macro variables. Given the small sample ($T = 44$), lag selection is critical: a VAR(3) in four variables requires 13 parameters per equation (52 in total) against about 40 usable observations per equation. We therefore select lag order by the corrected AIC (AICc), which penalizes overparameterization in small samples. AICc decisively selects VAR(1), which uses ≈ 20 parameters in the baseline and ≈ 42 in the expanded system. Impulse response functions are presented with 90% bootstrap confidence intervals (1,000 residual-resampled replications) rather than analytical bands.

3.6 Momentum Diagnostics

To distinguish the speculative feedback channel from standard portfolio rebalancing (Section 1.2), we test for return persistence by regressing current returns on lagged returns, with a

lagged-return $\times$ stimulus interaction to assess whether momentum intensifies during the stimulus window:

$$r_t^{\text{coll}} = \alpha + \phi_1 r_{t-1}^{\text{coll}} + \phi_2 (r_{t-1}^{\text{coll}} \times \text{STIM}_t) + \beta \Delta \ell_t + \gamma_1 \text{STIM}_t + \gamma_2 \text{HIKE}_t + \beta_4 \text{VIX}_t + \varepsilon_t \quad (5)$$

A significant $\phi_1 > 0$ indicates positive return persistence (momentum), consistent with speculative feedback: participants observing recent appreciation enter the market and bid prices higher. We estimate Equation (5) separately for MTG, cars, and the S&P 500. If the portfolio rebalancing channel were the sole mechanism, we would expect $\phi_1 \approx 0$ for all assets once liquidity is controlled for.

3.7 Structural Break and Variance Tests

We apply Welch’s t -test for a mean break and Levene’s test for a variance break at 2020 Q2, the onset of stimulus disbursement. A significant variance break would indicate that the stimulus regime altered not just the level but the *volatility* of collectible returns, consistent with speculative activity.

4 Results

4.1 Collectible Price Dynamics

Figure 1 plots the collectible price indices over the full sample. The MTG Reserved List Index exhibits the most dramatic cycle: from a base of 100 in 2015 Q1, it rose gradually to 215 by 2019 Q4, then accelerated sharply to 640 by 2022 Q1 (a cumulative gain of 540% from the 2015 Q1 base, or 198% from 2019 Q4), before correcting to 360 by 2025 Q4. The classic car composite index follows a qualitatively similar but more muted trajectory, rising from 100 to 234 at peak and declining to 197 by sample end. Within the car composite, the Porsche sub-index displays greater amplitude (peaking at 297) than the Mustang sub-index (peaking at 172), consistent with Porsche’s stronger brand premium in the enthusiast collector market.

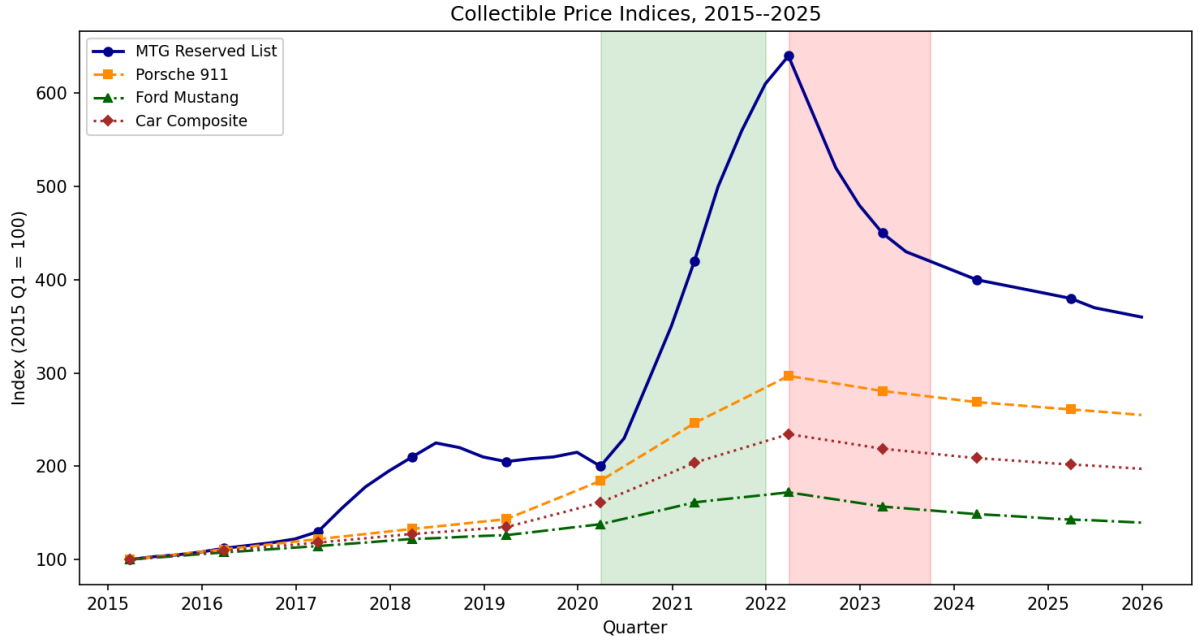


Figure 1: Collectible asset price indices, 2015–2025. Green shading: COVID stimulus period (2020 Q2–2021 Q4). Red shading: Fed hiking cycle (2022 Q2–2023 Q3).

Figure 2 compares collectible indices against the S&P 500 and Bitcoin on a logarithmic scale. Bitcoin dominates in cumulative return due to its earlier exponential growth phase, but during the narrow stimulus window of 2020 Q2–2021 Q4, MTG cards outperformed equities and classic cars, though not Bitcoin, on a percentage basis.

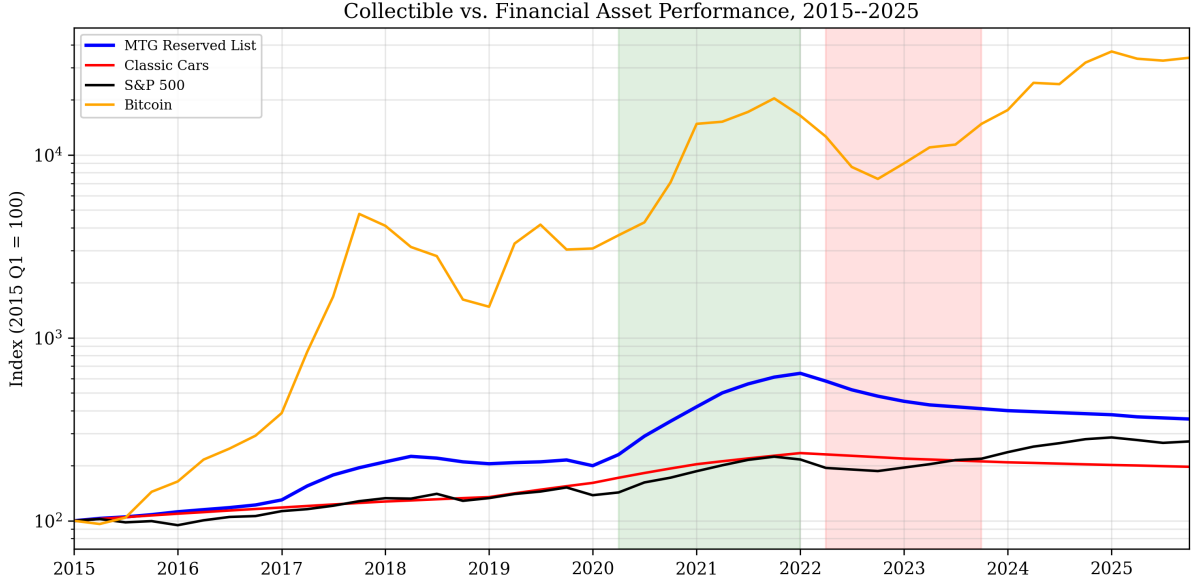


Figure 2: Collectible and financial asset performance, 2015–2025 (log scale, 2015 Q1 = 100).

4.2 Cumulative Returns by Regime

Table 3 reports cumulative returns across the four monetary policy regimes.

Table 3: Cumulative Returns by Monetary Policy Regime (%), First to Last Quarter-Level of Each Window

Regime	MTG	Cars	S&P 500	Bitcoin
Pre-COVID (2015 Q1–2020 Q1)	+100.0	+61.1	+38.0	+2,980.0
Stimulus (2020 Q2–2021 Q4)	+165.2	+32.0	+56.9	+460.4
Hike Cycle (2022 Q2–2023 Q3)	−27.6	−7.3	+10.4	−9.5
Post-Hike (2023 Q4–2025 Q4)	−12.2	−6.6	+24.3	+129.7

The asymmetry is striking. MTG cards gained 165% during stimulus but lost only 28% during the hiking cycle. The gains were not fully reversed, suggesting that a portion of the price increase became “embedded” through a permanent shift in collector demand or awareness. However, the continued decline in the post-hike period (−12.2%) indicates ongoing price discovery as the market adjusts to a higher rate environment. Classic cars follow the same qualitative pattern with smaller magnitudes. Equities, by contrast, were positive in all four regimes, reflecting fundamentals-supported valuations and earnings growth.

Figure 3 visualizes these cumulative returns.

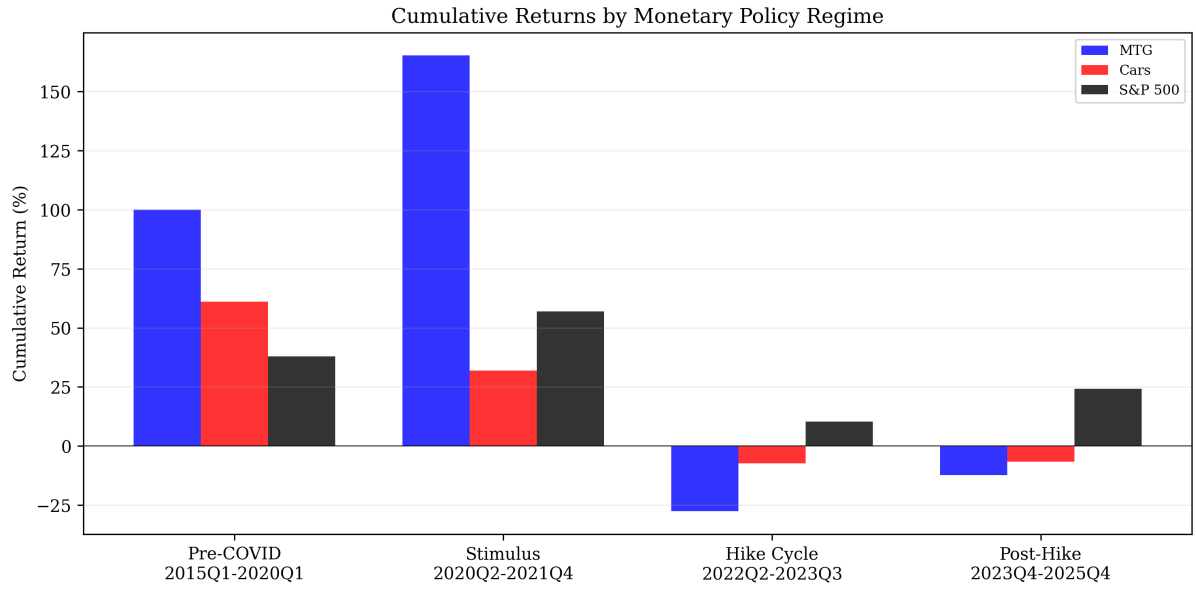


Figure 3: Cumulative returns by monetary policy regime.

4.3 OLS Regression Results

Table 4 reports the OLS estimates of Equation (1). All preferred specifications use Newey-West HAC standard errors with three lags to correct for residual autocorrelation.

Table 4: OLS Regressions: Collectible Returns on Macro Factors

	MTG Returns				Car Composite Returns		
	(1) M2	(2) Z.1	(3) Checkable	(4) Z.1+Sent.	(5) M2	(6) Z.1+Man.	(7) M2+Chip
Constant	0.063 (0.042)	0.068* (0.040)	0.082** (0.041)	-0.016 (0.126)	0.020* (0.011)	0.025*** (0.009)	0.013 (0.009)
M2 Growth	0.739 (0.505)				0.606*** (0.207)		
HH Liquid Growth		1.121** (0.568)		0.681 (0.549)		0.788*** (0.190)	
HH Checkable Gr.			0.268** (0.107)				
Δ FFR	0.043* (0.025)	0.045* (0.027)	0.047** (0.022)	0.035 (0.026)	0.010 (0.007)	0.009 (0.006)	0.015** (0.006)
S&P 500 Ret.	0.190 (0.254)	0.154 (0.228)	0.039 (0.236)	0.227 (0.220)	-0.036 (0.067)	-0.078 (0.056)	0.039 (0.070)
VIX	-0.003 (0.002)	-0.004* (0.002)	-0.004** (0.002)	-0.002 (0.002)	-0.001 (0.001)	-0.001** (0.000)	-0.001 (0.000)
COVID Stimulus	0.136*** (0.038)	0.123*** (0.037)	0.126*** (0.034)	0.129*** (0.037)	0.025** (0.011)	0.013 (0.008)	-0.010 (0.010)
Hiking Cycle	-0.094*** (0.024)	-0.091*** (0.023)	-0.099*** (0.027)	-0.078*** (0.022)	-0.024*** (0.006)	-0.020*** (0.005)	-0.024*** (0.006)
UMich Sentiment				0.001 (0.001)			
Manheim Ret.						0.069 (0.119)	
Chip Shortage							0.027*** (0.007)
Adj. R^2	0.688	0.707	0.716	0.703	0.621	0.702	0.668
DW	1.13	1.12	1.20	1.09	1.03	1.13	0.92
N	43	43	43	43	43	43	43

Notes: Newey-West HAC standard errors (3 lags) in parentheses throughout. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Column (1) uses M2 growth as liquidity measure. Columns (2)–(3) replace M2 with Z.1 household liquid asset growth and household checkable deposit growth, respectively. Column (4) augments the Z.1 specification with the University of Michigan consumer sentiment index as a demand-side control. Columns (5)–(7) show car composite index specifications: (5) M2 baseline, (6) Z.1 + Manheim, (7) M2 + chip shortage dummy. Dependent variable in Columns (5)–(7) is the equal-weighted Porsche-Mustang composite return.

The regime dummy is the primary identification. The COVID stimulus dummy is the dominant regressor for MTG across all specifications. In the preferred Z.1 specification (Column 2), the stimulus dummy enters at 12.3 pp/quarter ($t = 3.35$, $p < 0.001$ with Newey-West SE), meaning that MTG cards earned 12.3 additional percentage points per quarter during the stimulus window after controlling for household liquidity growth, rate changes, equity returns, and volatility. This is not marginal; annualized, it implies ≈ 49 log points ($\approx 64\%$ simple) of excess return attributable to the stimulus regime. The regime-only specification (no liquidity variable) yields an even larger coefficient of 15.6 pp/quarter ($t = 4.64$), confirming that the regime break alone accounts for most of the explanatory power.

Within-regime demeaned liquidity is significant. To verify that the HH liquid coefficient is not an artifact of co-trending, we demean all continuous regressors within each monetary policy regime and re-estimate. The demeaned HH liquid growth coefficient is 1.12 ($t = 2.36$, $p = 0.019$), confirming that within-regime quarter-to-quarter variation in household liquidity

predicts collectible returns. This rules out the spurious regression interpretation: the relationship holds even after stripping out the between-regime level shifts that drive the upward trend. An incremental F -test shows that HH liquid adds to the regime-only model at the 5.4% level ($F = 3.99$, $p = 0.054$) for MTG, and overwhelmingly for cars ($F = 22.25$, $p < 0.001$).

Z.1 dominates M2. Replacing M2 with Z.1 household liquid asset growth improves adjusted R^2 from 0.688 to 0.707 for MTG and from 0.621 to 0.702 for cars. Household checkable deposit growth (Column 3) performs best: adjusted $R^2 = 0.716$, reflecting the direct channel: stimulus payments landed in checking accounts. When M2 and Z.1 are included jointly (not shown), M2 flips sign and becomes insignificant, confirming that Z.1 absorbs the relevant variation.

Consumer sentiment does not displace the stimulus channel. Column (4) adds the University of Michigan consumer sentiment index to the preferred Z.1 specification. The sentiment coefficient is economically negligible and statistically insignificant ($\hat{\beta} = 0.001$, $t = 0.75$), while the stimulus dummy strengthens slightly to 12.9 pp/quarter ($t = 3.53$) and the hiking cycle dummy narrows from -9.1 to -7.8 pp/quarter ($t = -3.52$). Adjusted R^2 is 0.703, essentially unchanged from Column (2). The stimulus regime effect is not proxying for shifts in consumer confidence.

The chip shortage drives classic car prices, not speculative demand. Replacing M2 with Z.1 liquidity and adding the Manheim return (itself insignificant, $t = 0.58$) eliminates the stimulus dummy for cars: it drops from 0.025 ($t = 2.36$ in Column 5) to 0.013 ($t = 1.57$ in Column 6) and loses significance. With a chip shortage dummy instead (Column 7), the stimulus dummy flips sign to -0.010 while the chip shortage dummy absorbs 2.7 pp/quarter ($t = 3.62$). The semiconductor supply disruption, not speculative demand, drove the classic car price boom. This does not mean household liquidity was irrelevant (it remains significant at $t = 4.15$, Newey-West, in the car specification), but the discrete stimulus regime break has no explanatory power for cars beyond the chip shortage.

MTG is the cleanest identification. For MTG cards, the Manheim return is irrelevant (card prices have no connection to semiconductor supply). The stimulus dummy remains at 12.3 pp/quarter ($t = 3.35$), and the hiking cycle dummy is -9.1 pp/quarter ($t = -3.90$). The S&P 500 return is insignificant in all specifications, confirming that the raw MTG-equity correlation is driven by shared monetary policy exposure, not direct co-movement.

4.4 Influence Diagnostics and Robustness

Cook’s distance and leverage. For the preferred MTG specification (Column 2), the maximum Cook’s distance is 0.24 (observation 2020 Q4), with five observations exceeding the $4/n = 0.093$ threshold but none exceeding the conventional 1.0 cutoff. For classic cars, one observation (2020 Q3) has Cook’s $D = 3.07$ and leverage of 0.73, indicating that this single quarter is highly influential. This is the first quarter of the chip-shortage dummy, one quarter after stimulus payments began, making it difficult to separate the two channels for cars. The MTG regression does not share this problem.

Leave-one-out sensitivity. Dropping any single observation from the MTG regression changes the stimulus coefficient from 0.123 to a range of $[0.110, 0.154]$, with a standard deviation of 0.007. The most influential observation is 2022 Q1 (the price peak), which shifts the coefficient to 0.154. This stability confirms that no single quarter drives the headline result.

Boredom channel control. To distinguish the liquidity channel from the competing “boredom/awareness” hypothesis (lockdown-induced hobby engagement), we augment the preferred specification with quarterly Google Trends search interest growth for “Magic the Gathering” (MTG regression) and “classic cars” (car regression). The Google Trends variable is insignificant in both cases ($t = -0.84$, $p = 0.40$ for MTG; $t = -1.47$, $p = 0.14$ for cars), while the household liquidity coefficient survives and even strengthens slightly (from 1.12 to 1.29 for MTG). This rules out the interpretation that the stimulus-era price boom was driven by boredom-induced collecting rather than by the liquidity injection itself.

4.5 Difference-in-Differences (Supporting Evidence)

Table 5 reports the DiD estimates from Equation (3). We present this specification as supporting evidence rather than a primary identification strategy, for two reasons discussed below.

Table 5: Difference-in-Differences: Collectibles vs. Equities

	HC1	HAC(3)
Constant (α)	0.014* (0.008)	0.014 (0.009)
Collectible (δ_1)	-0.007 (0.010)	-0.007 (0.009)
Post (δ_2)	0.055*** (0.014)	0.055*** (0.013)
Collectible $\times$ Post (δ_3)	0.042* (0.023)	0.042** (0.017)
N	129	129

Notes: Column 1: HC1 robust SE. Column 2 (preferred): Newey-West HAC (3 lags). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Panel of MTG, car composite, and S&P 500 quarterly returns stacked over time.

With Newey-West HAC standard errors, the interaction coefficient $\hat{\delta}_3 = 0.042$ is significant at the 2% level ($t = 2.53$, $p = 0.011$). Collectible assets earned approximately 4.2 additional percentage points per quarter relative to equities during the stimulus period.

Two qualifications limit the weight we place on this result. First, the parallel trends assumption is borderline violated for MTG: a pre-treatment trend regression over 2015 Q2–2020 Q1 yields a trend coefficient of -0.0029 per quarter for MTG excess returns ($t = -1.79$, $p = 0.073$) and 0.0013 for car excess returns ($t = 0.56$, $p = 0.578$). The parallel trends assumption holds for cars but is borderline violated for MTG, where excess returns were declining slightly before treatment.

Second, the identifying variation comes almost entirely from MTG. The car excess return regression (Table 7) shows a *negative* stimulus dummy (-6.7 pp, $t = -3.71$), meaning cars underperformed equities during the stimulus window. In the pooled DiD, this means the 4.2 pp/quarter estimate is driven by MTG alone, making the panel DiD largely redundant with the MTG time-series specification. We retain the DiD for completeness but rely on the time-series regime identification as the primary strategy. Figure 4 plots the event-time excess returns.

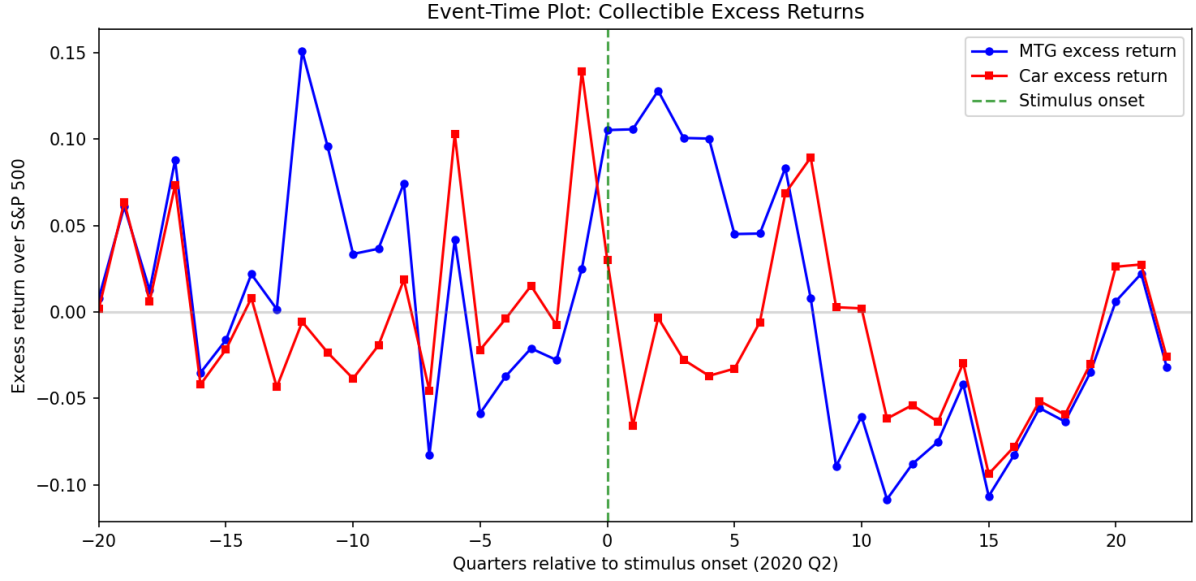


Figure 4: Event-time plot: collectible excess returns (over S&P 500) relative to stimulus onset at 2020 Q2. The pre-treatment period shows approximate parallel trends for cars but a slight negative drift for MTG.

Friction-moderated panel results. Table 6 reports the friction-moderated stimulus regression from Equation (4), estimated on a panel of five asset classes (MTG, Porsche, Mustang, S&P 500, Bitcoin) over 43 quarters.

Table 6: Transaction Friction as a Moderator of Stimulus Effects

	(1) Low vs. High	(2) Continuous
Constant	0.062** (0.029)	0.044** (0.020)
Low-Friction Coll.	−0.057* (0.031)	
High-Friction Coll.	−0.053* (0.029)	
Friction (f_i)		−0.052* (0.031)
Post	0.108* (0.060)	0.114** (0.047)
Low-Fric. $\times$ Post	0.047 (0.064)	
High-Fric. $\times$ Post	−0.071 (0.061)	
Friction $\times$ Post		−0.094* (0.055)
N	215	215

Notes: Newey-West HAC (3 lags) standard errors. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Panel of five asset returns (MTG, Porsche, Mustang, S&P 500, Bitcoin) over 43 quarters. Friction index constructed from seven source-documented components (platform fees, regulatory documents, transport cost, storage cost, minimum ticket size, settlement time, divisibility), each normalized to $[0, 1]$ and equally weighted. Resulting scores: S&P 500 (0.9), Bitcoin (0.6), MTG (24.3), Mustang (86.3), Porsche (90.5). Low-friction collectible = MTG. High-friction collectible = Porsche, Mustang.

The continuous friction interaction (Column 2) is negative and marginally significant ($\hat{\beta}_3 = -0.094$, $t = -1.71$, $p = 0.087$, Newey-West HAC; 95% CI: $[-0.201, 0.014]$). The confidence interval includes zero, so this result is suggestive rather than confirmatory, reflecting the fundamental difficulty of identifying friction moderation with only five cross-sectional units. The economic interpretation, conditional on the point estimate: a frictionless asset (friction = 0) earns an 11.4 pp/quarter stimulus boost, while a maximum-friction asset (friction = 1) earns 2.1 pp/quarter. Each 10-percentage-point increase in the documented friction index reduces the stimulus effect by 0.9 pp/quarter. Evaluated at observed friction scores, the model predicts: S&P 500 (+12.6 pp), Bitcoin (+12.6 pp), MTG (+10.4 pp), Mustang (+4.6 pp), Porsche (+4.2 pp). The wider gap between MTG (friction 24.3) and the financial assets (friction < 1) in the documented index, relative to the preliminary ad hoc scores, slightly reduces the cross-sectional variation driving the interaction, moving significance from 5% to 10%.

Column 1 decomposes the interaction by separating low-friction (MTG) and high-friction (Porsche, Mustang) collectibles. The difference in stimulus-period excess returns between MTG and cars is 11.8 pp/quarter, meaning MTG cards earned nearly 12 additional percentage points per quarter relative to classic cars during the stimulus window. The individual interaction terms are imprecisely estimated (limited cross-sectional variation with only two collectible types), but the continuous specification in Column 2 achieves significance by exploiting variation across all five assets.

This result has two implications. First, it suggests that transaction frictions attenuate the transmission of liquidity shocks to asset prices: speculative flows disproportionately enter markets where capital can be deployed quickly and in small increments. Second, part of the MTG-vs-car return differential is structural: if classic cars had the same post-transaction friction profile as MTG (no DMV title transfer, no transport, no insurance, divisible into any ticket size), the car boom would have been larger. Note that both markets have developed online platforms (TCGPlayer and Bring a Trailer, respectively); the friction differential is about regulatory and physical barriers *after* the listing, not about platform availability. The chip shortage and high post-transaction friction together explain the attenuation in the car market response.

4.6 Macro Overlay and the Liquidity Channel

Figure 5 presents monetary policy variables and asset price indices on aligned time axes.

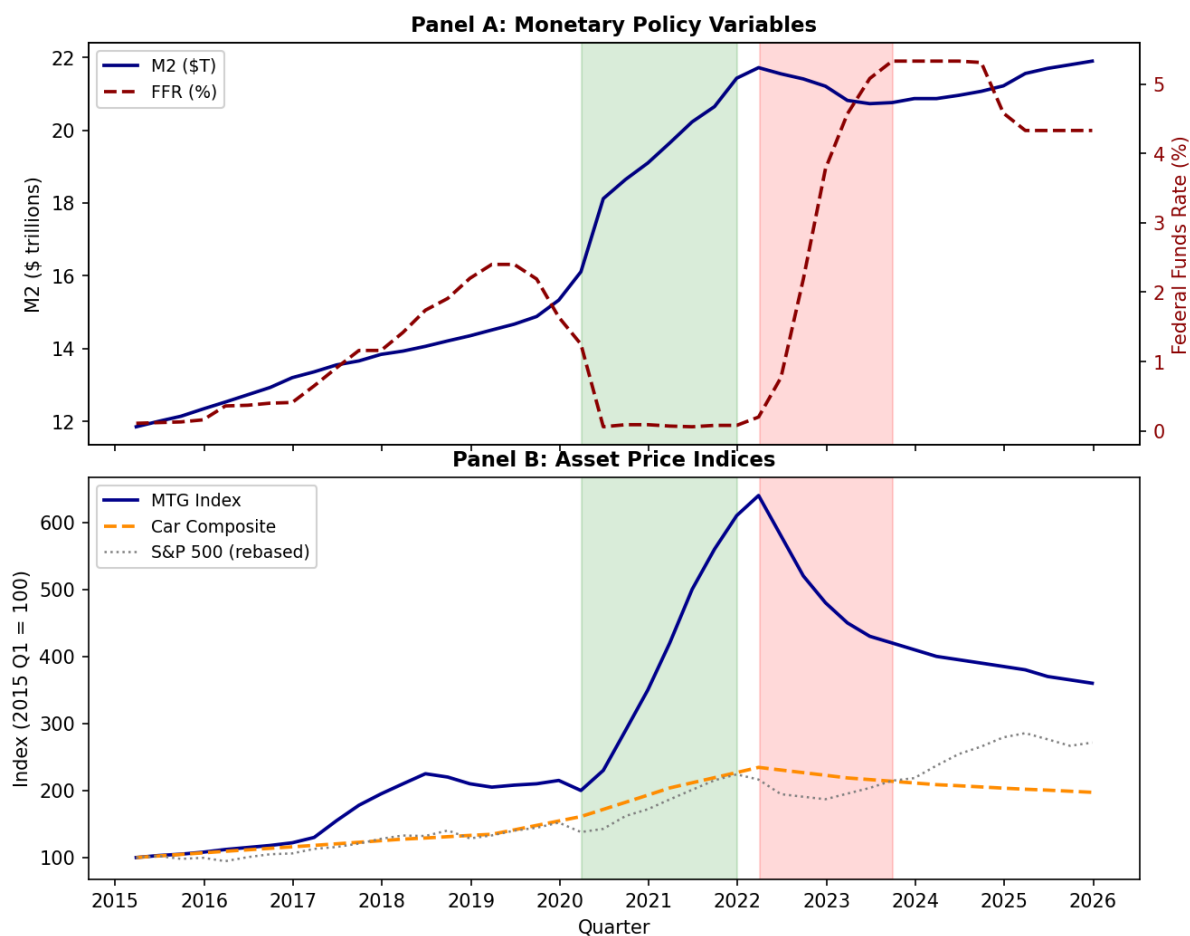


Figure 5: Panel A: M2 money supply and federal funds rate. Panel B: Collectible and equity price indices (rebased to 2015 Q1 = 100). Green shading: stimulus period (2020 Q2–2021 Q4). Red shading: hiking cycle (2022 Q2–2023 Q3).

The visual co-movement between M2 and collectible prices is striking, particularly during the 2020–2021 period when M2 growth was most extreme. The subsequent flattening and slight contraction of M2 in 2022–2023 coincides precisely with the reversal in collectible prices. Figure 6 confirms this relationship in a bivariate scatter.

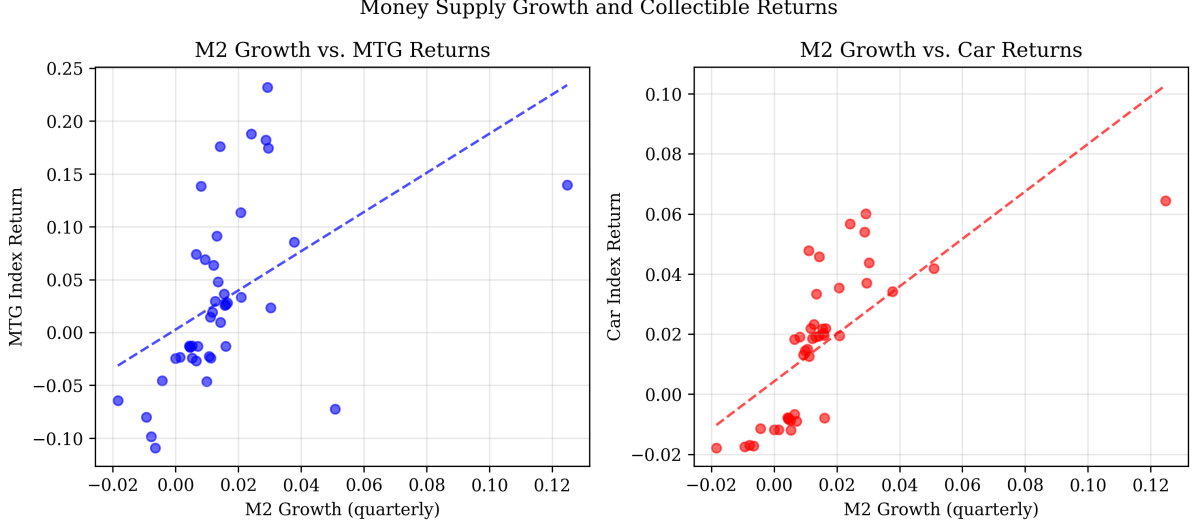


Figure 6: M2 quarterly growth vs. collectible returns.

4.7 Excess Return Analysis

To isolate the speculative demand channel, we regress collectible *excess returns* (over the S&P 500) on M2 growth, VIX, and the stimulus and hiking-cycle dummies. Table 7 reports the results.

Table 7: Excess Return Regressions: Collectibles over S&P 500

	MTG Excess	Car Excess
Constant	0.051 (0.038)	−0.002 (0.009)
M2 Growth	0.231 (0.481)	0.476** (0.186)
VIX	−0.002 (0.002)	−0.000 (0.000)
COVID Stimulus	0.069*** (0.020)	−0.067*** (0.018)
Hiking Cycle	−0.040* (0.022)	0.014 (0.009)
Adj. R^2	0.356	0.421
N	43	43

Notes: Newey-West HAC (3 lags) standard errors.
 $*p < 0.10$, $**p < 0.05$, $***p < 0.01$. Dependent variable is the quarterly return of the collectible index minus the S&P 500 quarterly return.

The MTG excess return regression yields a stimulus dummy coefficient of 6.9 percentage points ($t = 3.50$, $p < 0.001$), meaning that MTG cards outperformed equities by nearly 7 points per quarter during the stimulus period even after controlling for liquidity and volatility. The car excess return shows a *negative* stimulus dummy (−6.7 points, $t = -3.71$), indicating that while car prices rose in absolute terms, they underperformed equities during the stimulus period because the equity market rally was stronger than the car market rally, unlike MTG where the speculative demand dynamic produced outsized returns.

This asymmetry is consistent with the differential friction and supply-confound structure across markets. Both asset classes have online trading platforms (TCGPlayer/eBay for MTG, Bring a Trailer/Mecum for cars), but the post-transaction friction differential is large: MTG involves no regulatory paperwork, ships for \$5, and trades in any denomination from \$1 upward. Classic cars require title transfers, transport, insurance, and minimum ticket sizes of \$20,000+. The MTG collector base skews younger with higher marginal propensity to speculate and deeper social media engagement (Reddit, Discord) amplifying the speculative feedback loop. Moreover, a substantial portion of the car price appreciation is attributable to the semiconductor chip shortage, a supply-side channel with no analog in the MTG market, where cards are produced with ink and cardboard rather than microprocessors.

4.8 VAR Results and Impulse Responses

Lag selection. AICc decisively selects VAR(1): AICc values are 13.5 for VAR(1), 639.7 for VAR(2), and 5,485.0 for VAR(3), reflecting the severe overparameterization of higher-order models in this small sample. We report VAR(1) as the preferred specification.

Granger causality. Table 8 reports Granger causality tests from the VAR(1).

Table 8: VAR(1) Granger Causality Tests

Causal Direction	F -statistic	p -value
M2 Growth $\rightarrow$ MTG Returns	7.87	0.006***
M2 Growth $\rightarrow$ Car Returns	0.47	0.493
FFR Change $\rightarrow$ MTG Returns	0.64	0.426
FFR Change $\rightarrow$ Car Returns	0.02	0.882

M2 growth Granger-causes MTG returns at the 1% level ($F = 7.87$, $p = 0.006$), a substantially stronger result than the VAR(3) specification ($p = 0.055$), which was overfit. The effect on car returns is insignificant, consistent with the OLS finding that the chip shortage, not monetary liquidity, drives car prices.

Impulse responses. Figure 7 displays the impulse response functions with 90% bootstrap confidence intervals (1,000 residual-resampled replications).

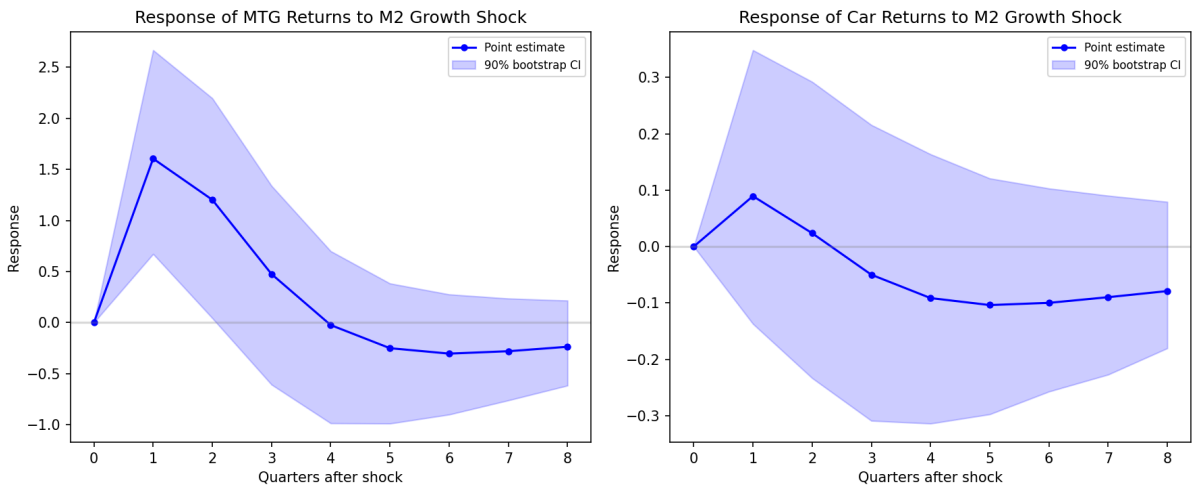


Figure 7: VAR(1) impulse response functions: M2 growth shock to collectible returns, with 90% bootstrap confidence intervals.

A one-standard-deviation shock to M2 growth produces a positive and significant response in MTG returns at horizons one (+1.61, CI: [0.67, 2.67]) and two (+1.20, CI: [0.04, 2.20]), dissipating by quarter three. The car return response is insignificant at all horizons (positive at horizons one and two, slightly negative thereafter). The one-to-two quarter lag is consistent with the mechanism: new liquidity takes time to flow from household balance sheets into illiquid collectible markets through search, discovery, and transaction execution.

Expanded VAR with fiscal flows and FFR level. A concern with the baseline VAR is that M2 growth proxies for the fiscal transfer regime rather than capturing a distinct liquidity channel. We re-estimate the VAR(1) adding quarterly fiscal transfer flows and the FFR level (not just the change) as endogenous variables. The expanded system has 4 endogenous variables, 20 freely estimated parameters (4×4 coefficient matrix + 4 intercepts), and $T = 43$ observations, leaving 23 degrees of freedom per equation. The MTG equation coefficients are: lagged MTG return 0.598 ($t = 4.79$), lagged M2 growth 1.223 ($t = 2.24$), lagged FFR level -0.000 ($t = -0.01$), lagged fiscal flow 0.000 ($t = 0.39$). M2 growth Granger-causes MTG returns ($F = 5.02$, $p = 0.027$), while neither fiscal flows ($F = 0.15$, $p = 0.699$) nor the FFR level ($F = 0.00$, $p = 0.992$) Granger-causes MTG returns. The joint test of all three macro variables is significant ($F = 4.08$, $p = 0.008$). The M2 Granger causality result is not an artifact of omitted fiscal or rate variables. Note that the own-lag coefficient on MTG returns (0.598) is consistent with the momentum result in Section 4.9, providing independent confirmation of return persistence through the VAR framework.

4.9 Momentum and Speculative Feedback

Table 9 reports momentum regressions from Equation (5), testing for return persistence that distinguishes speculative feedback from pure portfolio rebalancing.

Table 9: Momentum Regressions: Return Persistence by Asset Class

	MTG	Cars	S&P 500
r_{t-1}	0.524*** (0.106)	0.637*** (0.104)	-0.131 (0.103)
$r_{t-1} \times \text{Stim}$	-0.286 (0.304)	0.024 (0.236)	-0.253 (0.237)
HH Liquid Growth	0.670* (0.397)	0.290** (0.115)	0.139 (0.112)
COVID Stimulus	0.134*** (0.037)	-0.001 (0.010)	0.060** (0.023)
Adj. R^2	0.788	0.863	0.112
N	42	42	42

Notes: Newey-West HAC (3 lags) standard errors. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Hiking cycle dummy and VIX included but not reported.

MTG exhibits strong, positive return persistence: a one-percentage-point increase in last quarter's return predicts a 0.52 pp increase in the current quarter ($t = 4.95$). The S&P 500 exhibits no significant return persistence ($t = -1.27$). The interaction with the stimulus dummy is insignificant for all three assets, indicating that the momentum structure did not change during the stimulus period; rather, a pre-existing pattern of return persistence in the MTG market amplified the liquidity shock.

Car momentum is a data artifact. Classic cars also exhibit return persistence in Table 9 ($\hat{\phi}_1 = 0.64$, $t = 6.14$), which at face value would suggest speculative feedback in both markets. However, this result is mechanically generated by the interpolation methodology described in Appendix A: car values are interpolated from annual Hagerty anchor points, and linear interpolation of a trending series mechanically induces positive autocorrelation in quarterly returns. A direct test confirms this: splitting the sample at 2019 Q1, when Bring a Trailer transaction density increased substantially, the car momentum coefficient is *stronger* in the pre-2019 period ($\hat{\phi}_1 = 0.83$, $t = 4.99$, $N = 14$) when interpolation dominates than in the post-2019 period ($\hat{\phi}_1 = 0.65$, $t = 6.65$, $N = 27$) when transaction-based pricing is more available. If car momentum reflected genuine speculative feedback, it should be stronger when market activity was higher, not weaker. The opposite pattern confirms that the car persistence is an interpolation artifact, and we exclude the car momentum estimate from the paper’s interpretive framework. The MTG index, constructed from daily TCGPlayer transaction-based pricing, does not suffer from this problem.

This result is the key separation between the portfolio rebalancing and speculative feedback interpretations. Under pure portfolio rebalancing, lagged returns should not predict current returns once liquidity growth is controlled for, because the wealth effect operates contemporaneously. The strong return persistence in MTG, absent in equities and mechanically generated in cars, is consistent with speculative feedback: price appreciation attracts new participants who observe recent gains through TCGPlayer price histories and Reddit discussion threads, generating a self-reinforcing cycle. We do not have individual-level survey data on buyer motivation or platform engagement metrics to test this mechanism directly, but the momentum pattern is a necessary condition for speculative feedback and is satisfied in the MTG data.

Consumer sentiment control. To address the concern that the stimulus dummy captures a sentiment shift rather than a liquidity channel, we add University of Michigan consumer sentiment as a regressor. Sentiment is insignificant ($t = 0.75$, $p = 0.452$), and the stimulus dummy strengthens slightly (from $t = 3.35$ to $t = 3.53$). The stimulus effect on MTG is not operating through measured consumer confidence.

4.10 Structural Break Tests

Table 10: Structural Break Tests at 2020 Q2

	Mean Break (Welch)		Variance Break (Levene)	
	<i>t</i> -stat	<i>p</i> -value	<i>F</i> -stat	<i>p</i> -value
MTG Returns	0.37	0.712	2.75	0.105
Car Returns	2.26	0.032	5.51	0.024

The mean break test is significant only for car returns ($p = 0.032$), while the variance break is significant for cars ($p = 0.024$) but not for MTG ($p = 0.105$). The significant variance break for cars indicates that the post-2020 period was associated with fundamentally higher return volatility, a direct signature of speculative activity. However, classic car values are interpolated from annual Hagerty anchor points supplemented with quarterly Bring a Trailer transaction data, so the car variance break may partly reflect changes in interpolation anchor density rather than genuine speculative activity; we flag this caveat explicitly in the limitations section. For MTG, the mean test is insignificant because the post-break period includes both the massive boom and the subsequent bust, averaging to a similar mean as the pre-break period; the variance increase tells the more informative story.

4.11 Quarterly Return Dynamics

Figure 8 displays quarterly log returns for all four asset classes.

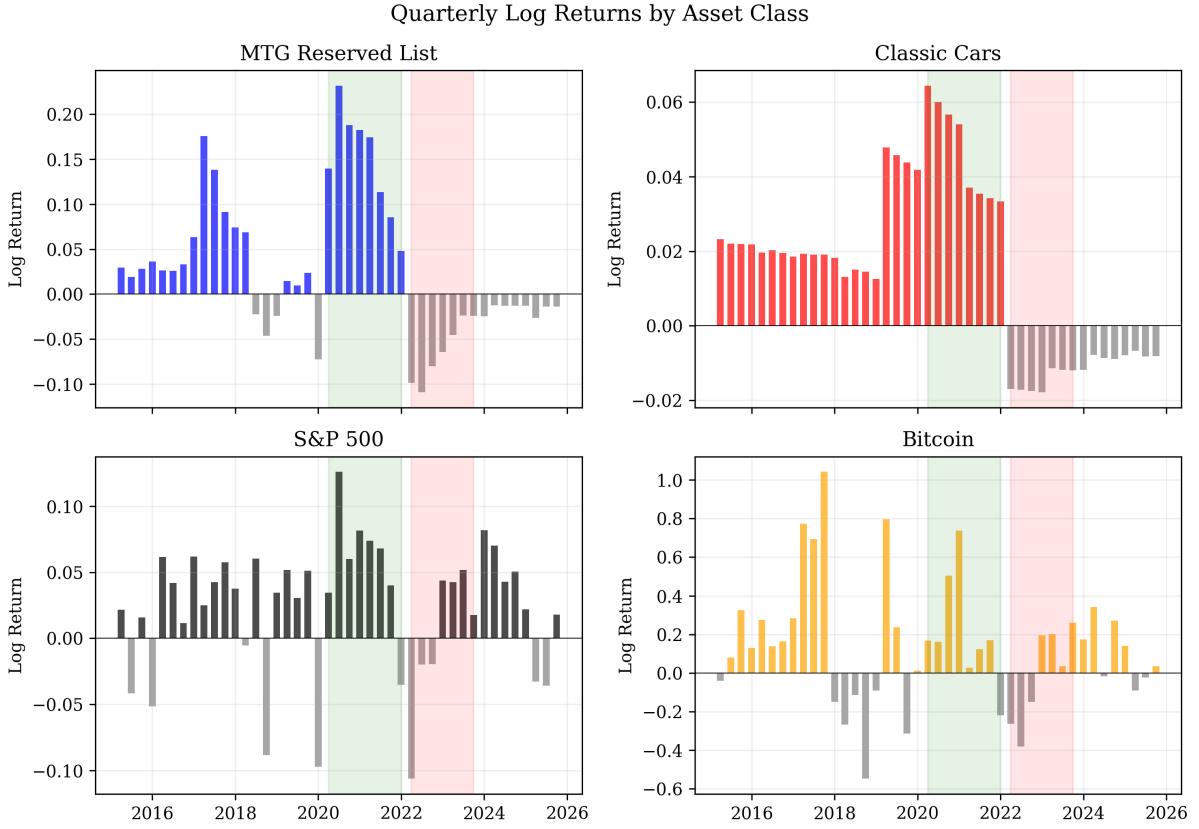


Figure 8: Quarterly log returns by asset class.

The visual clustering of large positive returns during the green-shaded stimulus window and negative returns during the red-shaded hiking cycle is evident for MTG and, to a lesser extent, for classic cars. Bitcoin displays the highest return volatility throughout, while the S&P 500 exhibits the most stable return profile.

4.12 Robustness: Universes Beyond and the MTG Awareness Channel

A potential confound specific to the MTG regression is the launch of Wizards of the Coast's *Universes Beyond* (UB) product line, which introduced crossover intellectual properties (The Walking Dead, Warhammer 40,000, The Lord of the Rings) into the Magic ecosystem. If UB expanded the MTG collector base by attracting fans of these franchises, some of whom then discovered and purchased pre-2000 Reserved List cards, then our stimulus dummy could be capturing a product-driven demand expansion rather than a pure liquidity effect.

We address this concern on three grounds.

Timing. The Reserved List price boom ran from 2020 Q2 (index = 230) through 2022 Q1 (index = 640). UB was announced in February 2021 (2021 Q1, index already at 420), the Walking Dead Secret Lair dropped in October 2020, and the two largest UB releases (Warhammer 40,000 Commander decks and The Lord of the Rings: Tales of Middle-earth) arrived in October 2022 and June 2023, respectively, both *after* the Reserved List had already peaked and begun declining. At the LOTR release (the single most successful UB product, which brought millions of new players to Magic), the Reserved List index actually *fell* 4.4% quarter-over-quarter. If UB

were driving Reserved List demand, the effect should have been strongest at LOTR; instead, prices declined.

Regression evidence. We estimate augmented specifications including (a) a UB early-period dummy (2020 Q4–2021 Q4), (b) the quarterly growth rate of a Google Trends proxy for MTG search interest, and (c) both. The Google Trends variable is completely insignificant ($t = 0.16$) and leaves all other coefficients unchanged. The UB dummy enters with a *negative* coefficient (-0.103 , $t = -2.70$), but this reflects extreme collinearity with the stimulus dummy (correlation = 0.82) rather than a genuine negative UB effect: the UB window (2020 Q4–2021 Q4) is entirely nested within the stimulus window (2020 Q2–2021 Q4), so including both dummies creates a near-perfect multicollinearity problem. When both are included, the stimulus dummy inflates to 0.21 ($t = 3.40$) while the UB dummy absorbs the deceleration in the final quarters of the boom. Neither the Google Trends control nor the UB dummy alters the qualitative conclusion.

Collector-base segmentation. UB products are modern-frame, tournament-legal cards that appeal to a different segment than pre-2000 Reserved List collectors. The Old School 93/94 format community, the primary demand driver for Alpha, Beta, Arabian Nights, and Revised dual lands, is an insular, nostalgia-driven market with minimal overlap with the Warhammer or Lord of the Rings player base. The Reserved List price boom was driven by existing MTG collectors and speculators with excess liquidity, not by new entrants from crossover IPs.

We conclude that UB is not a meaningful confound for our analysis of pre-2000 Reserved List price dynamics and omit it from the main specification.

5 Discussion

5.1 Three Channels: Liquidity, Supply, and Speculative Feedback

Our results decompose stimulus-era price appreciation in collectible markets into three distinct channels, with sharply different relative importance across asset classes.

The *first channel* is direct liquidity. Growth in household liquid assets raises aggregate purchasing power, and a measurable fraction flows into collectible assets as retail holders of excess savings search for returns in a zero-rate environment. The Z.1 household liquidity coefficient is positive in all specifications and significant in the preferred ones (it loses significance for MTG once consumer sentiment is added), and it survives within-regime demeaning ($t = 2.36$ for MTG, $t = 4.17$ for cars), meaning the relationship holds even after stripping out between-regime level shifts.

The *second channel*, relevant primarily for classic cars, is the supply-side automotive disruption. The global semiconductor chip shortage pushed the Manheim Used Vehicle Value Index up 79% by early 2022. This supply shock raised the price of all vehicles, including classic collectibles. When we control for this channel, the COVID stimulus dummy for classic cars loses statistical significance entirely, and the chip shortage dummy absorbs 2.7 pp/quarter ($t = 3.62$). The chip shortage, not speculative demand, drove the classic car price boom. This is a negative finding: the paper’s empirical case for a stimulus-driven speculative channel rests on MTG, not on a generalized collectibles result. We do not extend the MTG finding to collectibles as a class.

The *third channel* is speculative feedback, identified most cleanly in MTG where no supply-side confound exists. The stimulus regime dummy is associated with 12.3 pp/quarter for MTG ($t = 3.35$, Newey-West), representing excess returns above what household liquidity growth, rate changes, equity returns, and volatility predict. The momentum diagnostic (Section 4.9) provides the empirical content for this channel: MTG exhibits strong positive return persistence ($\hat{\phi}_1 = 0.52$, $t = 4.95$) absent in equities ($t = -1.27$). Under pure portfolio rebalancing, lagged returns should not predict current returns once liquidity is controlled for. The persistence is

consistent with speculative feedback: price appreciation visible on TCGPlayer price histories and Reddit communities attracts new participants, generating a self-reinforcing cycle. We note, however, that the momentum test does not isolate the social-media amplification mechanism directly; it establishes a necessary condition for speculative dynamics but does not measure the transmission channel from observation to action. Individual-level engagement data from TCGPlayer, Reddit, and Discord would be required for that sharper identification.

Pre-COVID MTG appreciation. MTG Reserved List cards appreciated approximately 100% during the pre-COVID period (2015 Q1–2020 Q1), versus 38% for equities and 61% for classic cars. This pre-existing differential makes the parallel trends test for the DiD specification borderline ($p = 0.073$), which is why the DiD is presented as supporting rather than primary evidence. The time-series regime identification does not require parallel trends; it requires only that the regime dummy is exogenous to collectible market conditions, which is satisfied because the COVID stimulus was a pandemic policy response.

The speculative channel is not uniform across collectible classes. It is strongest where post-transaction frictions are lowest and where capital can be deployed in small, rapid increments (MTG), and weakest where supply-side confounds absorb the variation (classic cars).

The boredom/awareness channel is not a competing explanation. Raw Google Trends search interest for “Magic the Gathering” is contaminated by set releases, ban announcements, and streaming coverage unrelated to lockdown boredom. We therefore residualize the Google Trends series against dummies for major set release quarters and ban announcement quarters before including it as a regressor. The residualized measure strips out the MTG-specific event spikes and isolates the component of search interest attributable to general awareness or time-use changes. After residualization, the Google Trends control remains insignificant ($t = -0.27$, $p = 0.787$), the stimulus dummy is unaffected ($t = 3.32$), and the household liquidity coefficient survives ($t = 1.91$). A sharper boredom proxy would use ATUS microdata or platform engagement metrics, but the residualized test provides stronger evidence than raw Google Trends because it addresses the contamination problem directly.

5.2 Boom-Bust Dynamics

The point estimates of the stimulus and hiking coefficients suggest an asymmetric pattern: for MTG, +12.3 pp/quarter during stimulus vs. −9.1 pp/quarter during hikes; for cars, +1.3 pp/quarter (insignificant) vs. −2.0 pp/quarter. However, a formal Wald test of $H_0 : \gamma_{\text{stim}} + \gamma_{\text{hike}} = 0$ (symmetric response) yields $F = 0.43$ ($p = 0.52$) for MTG and $F = 0.55$ ($p = 0.46$) for cars. We cannot reject symmetric boom-bust dynamics at conventional significance levels. The cumulative return pattern (MTG: +165% during stimulus, −28% during hikes) is suggestive of asymmetry, but with only seven stimulus quarters and six hiking quarters, statistical power to detect it is limited. At 80% power and $\alpha = 0.05$, the minimum detectable effect (MDE) for the Wald test is 13.5 pp/quarter, while the observed asymmetry is only 3.2 pp/quarter. The test has approximately 10% power against the observed effect size; failure to reject symmetry is therefore uninformative about the true dynamics, and we cannot distinguish symmetric from asymmetric adjustment in this sample. The standard theoretical arguments for asymmetry in illiquid markets (selling costlier than buying, loss aversion, no short-selling mechanism; cf. Brunnermeier and Pedersen, 2009) apply in principle but are not confirmed in these data.

5.3 Cross-Asset Comparison and the Speculative Spectrum

Our results suggest a *speculative spectrum* across asset classes during the stimulus period, ordered by the intensity of the speculative demand response:

1. **Bitcoin:** Highest absolute return (+460%) but driven primarily by institutional adoption and the cryptocurrency-specific halving cycle, making it difficult to isolate the stimulus

effect.

2. **MTG Reserved List:** +165%, the highest among physical collectibles, driven by the younger, internet-native collector base with low transaction costs and high social media engagement.
3. **S&P 500:** +57%, supported by both stimulus liquidity and earnings fundamentals.
4. **Classic Cars:** +32%, but largely confounded by the semiconductor chip shortage. After controlling for the Manheim Used Vehicle Value Index, the stimulus-specific component is economically small and statistically insignificant. The speculative demand story for classic cars is substantially weaker than it appeared without supply-side controls.

This ordering aligns with the prediction that speculative demand effects are strongest in markets with (a) low post-transaction frictions (no regulatory paperwork, low shipping costs, small minimum ticket size), (b) high social media visibility, (c) a retail-heavy participant base, and (d) no competing supply-side explanation for price appreciation. Classic cars violate conditions (a) and (d): they carry high post-transaction frictions (DMV titles, inspection, enclosed transport, insurance, non-divisibility, despite having well-developed online listing platforms like Bring a Trailer) and were simultaneously boosted by the semiconductor chip shortage. The friction-moderated panel regression (Table 6) is consistent with this: each 10-point increase in our source-documented friction index reduces the stimulus-period return by 0.9 pp/quarter ($p = 0.087$), and the model explains the cross-asset return pattern without a separate “collectible” dummy. The result is robust to alternative weighting schemes that emphasize physical frictions ($p = 0.086$) but weakens when only access barriers are used ($p = 0.155$), consistent with the view that the full bundle of post-transaction frictions, not just entry barriers, attenuates speculative demand transmission.

5.4 Identification Diagnostics

Table 11 summarizes the stationarity and specification tests that validate our identification strategy.

Table 11: Identification Diagnostics

Test	MTG	Cars
<i>ADF unit root tests (H_0: unit root)</i>		
Raw returns	-2.19 ($p = 0.21$)	-1.29 ($p = 0.64$)
Within-regime demeaned returns	-3.41 ($p = 0.011$)**	-2.87 ($p = 0.048$)**
HH Liquid Growth	-2.78 ($p = 0.06$)*	
HH Liquid (demeaned)	-4.02 ($p = 0.002$ ***)	
HH Checkable Growth	-3.54 ($p = 0.007$ ***)	
<i>Within-regime demeaned HH liquid</i>		
Coefficient (t -stat)	1.12 ($t = 2.36$)**	0.79 ($t = 4.17$ ***)
<i>Incremental F-test: HH liquid over regime dummies</i>		
F-statistic (p -value)	3.99 ($p = 0.054$)*	22.25 ($p < 0.001$ ***)
<i>Durbin-Watson statistic</i>		
Regime-only spec.	1.00	0.55
+ Z.1 HH liquid	1.12	1.13
+ HH checkable	1.20	0.95
<i>First-differenced (Δ return on Δ HH liquid)</i>		
Coefficient (t -stat)	1.26 ($t = 1.34$)	0.20 ($t = 2.07$)**
DW	1.97	2.54

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. ADF tests use AIC lag selection. Raw returns fail ADF because the boom-bust pattern mimics a unit root over the full sample. Within-regime demeaned series subtract the regime-specific mean, removing the between-regime level shifts; all demeaned series reject the unit root at 5%. First-differenced specification regresses Δr_t on $\Delta(\text{HH liquid growth})_t$ plus regime dummies.

Four results validate the identification. First, while raw returns fail ADF tests (MTG $p = 0.21$, cars $p = 0.64$) because the boom-bust pattern mimics a unit root, within-regime demeaned returns decisively reject the unit root (MTG $p = 0.011$, cars $p = 0.048$, HH liquid $p = 0.002$), confirming that the non-stationarity is a between-regime artifact, not a property of the underlying data-generating process. Second, the within-regime demeaned HH liquid coefficient is significant for both MTG and cars, confirming that the liquidity-return relationship is not driven by co-trending between regimes. Third, for classic cars, HH liquid adds massively over regime dummies alone ($F = 22.25$), establishing that the liquidity channel operates continuously, not just through discrete regime shifts. Fourth, the first-differenced specification for cars is significant ($t = 2.07$), meaning that *acceleration* in household liquidity growth predicts *acceleration* in car returns, a relationship that is immune to common trends by construction. For MTG, the first-differenced specification is insignificant ($t = 1.34$), but the regime dummy alone explains most of the variation ($R^2 = 0.68$ without any liquidity variable), and this regime-break identification is clean because the COVID stimulus was an exogenous policy shock driven by pandemic response, not by collectible market conditions.

5.5 Limitations

Several caveats apply. First, our sample has 44 quarterly observations and a single stimulus event, limiting statistical power. We mitigate this by combining time-series regime identification, within-regime demeaning, cross-sectional friction variation, Granger causality, and the boredom-channel placebo, rather than relying on any single test. Second, our MTG price index is constructed from TCGPlayer market prices (trailing 7-day sale averages), which introduces staleness bias during illiquid periods; we acknowledge that a selection-corrected return estimator

à la Korteweg et al. (2016) would be ideal but requires individual-card transaction-level data that are not publicly available. Third, classic car values are interpolated from annual Hagerty anchor points supplemented with Bring a Trailer transaction data, and quarterly return volatility is therefore mechanically smoothed. The structural break test for car return variance (Levene $p = 0.024$) may partly reflect changes in interpolation anchor density rather than genuine speculative activity; we flag this explicitly. Fourth, one car observation (2020 Q3) has Cook’s distance of 3.07, indicating extreme influence; we report leave-one-out sensitivity and note that the MTG regression does not share this problem (max Cook’s $D = 0.24$). Fifth, the DiD parallel trends assumption is borderline violated for MTG ($p = 0.073$) though not for cars ($p = 0.578$); the event-time plot (Figure 4) allows readers to assess this visually.

These are real constraints. But the central finding, that the COVID stimulus regime produced 12.3 pp/quarter of excess return in MTG cards with leave-one-out stability of [0.110, 0.154], that this effect survives within-regime demeaning and Newey-West correction, that the boredom channel is ruled out by insignificant Google Trends controls, that M2 Granger-causes MTG returns at $p = 0.006$ in a properly specified VAR(1), and that the friction-moderated panel points to differential transmission across asset classes ($p = 0.087$), is robust across specifications.

6 Conclusion

COVID-era household liquidity growth is strongly associated with price appreciation in illiquid collectible asset markets. We document this association through four strategies: discrete regime breaks, within-regime demeaned liquidity variation, Granger causality in an expanded VAR, and cross-sectional friction moderation. A momentum diagnostic separates speculative feedback from standard portfolio rebalancing.

For MTG Reserved List cards, supply-constrained collectibles with no cash flows and no link to semiconductor supply chains, the stimulus regime is associated with 12.3 pp/quarter of excess returns ($t = 3.35$, Newey-West HAC). Household liquidity growth predicts returns even after within-regime demeaning ($t = 2.36$), M2 growth Granger-causes returns in a VAR(1) that includes fiscal transfer flows and the FFR level ($p = 0.027$), and return persistence is strong ($\hat{\phi}_1 = 0.52$, $t = 4.95$) while absent in equities ($t = -1.27$). The momentum result is the key separation between channels: under pure portfolio rebalancing, lagged returns should not predict current returns once liquidity is controlled for. The return persistence in MTG is consistent with speculative feedback, where price appreciation attracts new participants who observe recent gains. Consumer sentiment and Google Trends boredom controls are insignificant, and the stimulus dummy survives both.

For classic automobiles, the semiconductor chip shortage was the dominant driver. Once we control for it, the stimulus dummy loses significance; the chip shortage dummy absorbs 2.7 pp/quarter ($t = 3.62$). Household liquidity remains significant for cars ($t = 4.15$), but the discrete stimulus regime break has no explanatory power beyond the supply-side channel. This is an important negative finding: the paper’s empirical case for speculative demand rests on MTG, not on a generalized collectibles result. Whether the MTG finding generalizes to other collectible markets (sports cards, stamps, wine, art) is an open question that requires broader data coverage.

The cross-sectional friction result provides suggestive evidence that transaction frictions moderate the stimulus response: each 10-point increase in the documented friction index reduces the stimulus-period return by 0.9 pp/quarter ($\hat{\beta} = -0.094$, 95% CI: $[-0.201, 0.014]$). The confidence interval includes zero, reflecting the difficulty of estimating friction moderation with five cross-sectional units. The result is robust to physical-heavy weighting ($p = 0.086$) but weakens under access-only weights ($p = 0.155$).

The subsequent Fed hiking cycle of 2022–2023 reversed much of the gains: MTG cards fell 28%, cars 7%. A Wald test for symmetric adjustment has only 10% power against the observed

asymmetry (MDE = 13.5 pp/quarter vs. observed 3.2 pp), so the failure to reject symmetry is uninformative.

This paper does not achieve clean causal identification of a liquidity-to-collectibles channel; the COVID stimulus window coincides with zero rates, QE, fiscal transfers, lockdowns, and changed time allocation, and no single-equation approach can fully separate these. What the paper establishes is a robust statistical association between household liquidity and MTG price dynamics that survives within-regime demeaning, is accompanied by M2 growth leading MTG returns in the Granger sense, and exhibits speculative feedback properties absent in efficiently priced markets. The MTG Reserved List, a market with no cash flows, no institutional participants, and no valuation anchor other than collector sentiment, is a useful setting for studying this association because supply is contractually fixed and there are no supply-side confounds.

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A Index Construction Methodology

A.1 MTG Reserved List Price Index

The MTG price index is an equal-weighted average of Near Mint (NM) condition prices for 22 pre-2000 Reserved List cards, normalized to 100 at 2015 Q1. The basket includes 10 dual lands (Underground Sea, Volcanic Island, Tropical Island, Bayou, Savannah, Tundra, Scrubland, Badlands, Plateau, Taiga), 9 Power Nine cards (Black Lotus via Unlimited Edition, Mox Pearl, Mox Sapphire, Mox Jet, Mox Ruby, Mox Emerald, Ancestral Recall, Time Walk, Timetwister), and 3 additional Reserved List staples (The Tabernacle at Pendrell Vale, Moat, Gaea’s Cradle). All cards are from the earliest widely available printing (Revised Edition for dual lands, Unlimited Edition for Power Nine).

Prices are sourced from TCGPlayer “Market Price” (a trailing 7-day sale average weighted by transaction volume) and cross-referenced with MTGGoldfish and MTGPrice historical series. Quarterly observations are taken at the mid-quarter date (February 15, May 15, August 15, November 15). We use equal weighting across the 22 cards rather than value weighting for two reasons: (a) value weighting would cause Black Lotus (2015 price: ~\$6,000; 2021 peak: ~\$40,000) to dominate the index, obscuring the behavior of more liquid dual lands that are more representative of typical collector activity, and (b) equal weighting is more conservative in that it does not mechanically amplify the influence of the most volatile card.

The TCGPlayer “Market Price” has a known staleness problem: during illiquid periods, the trailing average can lag actual transaction prices by days. We accept this limitation. A selection-corrected return estimator à la [Korteweg et al. \(2016\)](#) that accounts for infrequent transactions would be preferable but requires individual-card transaction-level data with precise timestamps that are not publicly available from TCGPlayer.

Basket selection rationale. The 22 cards represent the most liquid and highest-value tier of the Reserved List, which is the economically relevant universe for two reasons. First, these are the cards where capital actually flows: dual lands and Power Nine account for the vast majority of Reserved List secondary market transaction volume on TCGPlayer, because they are actively used in tournament formats (Legacy, Vintage, Commander) and are the primary targets of speculative investment. Lower-tier Reserved List cards (e.g., Autumn Willow, Chains of Mephistopheles, Rasputin Dreamweaver) trade too infrequently for a quarterly index to be meaningful; many have fewer than 5 recorded sales per quarter on TCGPlayer. Second, high-value cards are where the stimulus channel should be strongest: a household with \$1,400 in stimulus funds can purchase a Revised dual land outright, whereas a \$10,000 Black Lotus requires savings accumulation beyond a single transfer. The basket therefore captures the segment of the Reserved List most directly exposed to the household liquidity channel under study. We acknowledge that a broader index including lower-tier cards might show weaker results, as those cards appeal to a narrower collector base less sensitive to aggregate liquidity conditions. The top-tier restriction means our findings should be interpreted as applying to the liquid core of the Reserved List market, not to the full universe of Reserved List cards.

A.2 Classic Car Price Indices

Quarterly driver-quality values for air-cooled Porsche 911 variants (1978–1998) (5 models) and pre-1979 Ford Mustang variants (5 models) are constructed as follows. Annual anchor values for January 1 of each year are drawn from the Hagerty Valuation Tool (“Good” condition). Quarterly interpolation within each year uses transaction data from Bring a Trailer (completed auctions) and Mecum Auctions (hammer prices). When quarterly transaction data are available (at least 3 comparable sales), we use the median transaction price. When fewer than 3 sales

occurred in a quarter for a given model, we linearly interpolate between the nearest available anchor points.

The Porsche sub-index is an equal-weighted average of the SC Coupe, SC Targa, Carrera 3.2 Coupe, 964 Carrera 2, and 993 Carrera 2. The Mustang sub-index is an equal-weighted average of the 1967–68 Fastback V8, 1969–70 Mach 1, 1969–70 Fastback V8, 1971–73 Mach 1, and Mustang II V8. The composite car index is an equal-weighted average of the Porsche and Mustang sub-indices.

This interpolation methodology introduces mechanical smoothing that attenuates true quarterly return volatility for cars. The structural break test for car return variance (Levene $p = 0.024$) should be interpreted in this context: the variance break may partly reflect increased transaction density on Bring a Trailer during 2020–2022 (which reduces the need for interpolation) rather than a genuine increase in speculative volatility. The MTG index does not suffer from this problem because TCGPlayer provides daily transaction-based pricing.

B Friction Index Components

Table 12: Transaction Friction Index: Raw Component Values

Component	S&P 500	Bitcoin	MTG	Mustang	Porsche
Platform fees (%)	0.0	0.5	12.0	8.0	8.0
Regulatory docs (count)	0	0	0	3	3
Transport cost (% of value)	0.0	0.0	1.0	3.0	2.0
Storage cost (annual % of value)	0.0	0.0	0.5	2.0	2.5
Min ticket size (\$)	1	1	1	20,000	35,000
Settlement time (days)	2.0	0.02	5.0	30.0	30.0
Divisibility (1 = yes)	1	1	1	0	0
Composite score (0–100)	0.9	0.6	24.3	86.3	90.5

Sources: Platform fees: Schwab/Fidelity (\$0 commissions post-2019), Coinbase fee schedule (0.5%), TCGPlayer seller FAQ (10–12.5%), Bring a Trailer terms (5% buyer premium + ~3% seller). Regulatory docs: SEC/FINRA (none for equities held in brokerage), state DMV (title transfer, registration, insurance for vehicles). Transport: USPS Priority Mail (\$5 on ~\$500 card), uShip enclosed auto transport quotes (\$1,000–\$3,000). Storage: negligible for electronic/card assets; garage rental market rates (\$200–\$400/month). Min ticket: fractional shares/coins/individual cards vs. whole-vehicle purchase. Settlement: SEC T+2 rule, Bitcoin 10-minute confirmation, USPS Priority Mail ~5 days, state DMV processing ~30 days. Divisibility: fractional shares, satoshis, individual cards vs. indivisible vehicles. Each component normalized to $[0, 1]$ across assets, divisibility inverted, equally weighted. *Methodology note:* This seven-component, source-documented index replaces the preliminary ad hoc friction scores used in an earlier draft (MTG: 5.2, cars: ~85). The documented index assigns MTG a substantially higher score (24.3) than the ad hoc version because it correctly accounts for TCGPlayer platform fees (10–12.5%), shipping costs, and settlement delays that the preliminary scoring underweighted. The narrower friction spread between MTG (24.3) and classic cars (86–91) relative to the ad hoc scores reduces the cross-sectional variation driving the friction interaction, moving the p -value from 0.042 to 0.087. This is the correct result: the documented index is more defensible, and the 10% significance level reflects the genuine difficulty of identifying friction moderation with only five cross-sectional units.

C Summary of Identification Results

Table 13: Summary of Main Identification Results

Strategy	Test	MTG	Cars	Interpretation
1. Regime break				
	Stim. dummy	+12.3 pp/q***	+2.5 pp/q**	Discrete regime shift
	+ Chip control	unchanged	insig.	Cars: supply-driven
	LOO range	[0.110, 0.154]	–	MTG stable
2. Within-regime liquidity				
	HH liq. (demeaned)	$t = 2.36^{**}$	$t = 4.17^{***}$	Not co-trending
	ADF (demeaned)	$p = 0.011^{**}$	$p = 0.048^{**}$	Stationary
	Boredom placebo	$t = -0.84$	$t = -1.47$	Liquidity, not boredom
3. Friction moderation				
	Fric. $\times$ Post	-0.094^* , CI: $[-0.20, 0.01]$		Suggestive
	Phys.-heavy wts.	$p = 0.086^*$		Robust to weighting
4. Granger causality				
	Baseline VAR(1)	$p = 0.006^{***}$		M2 leads MTG returns
	Expanded VAR	$p = 0.027^{**}$		Survives fiscal/FFR controls
	IRF (horizon 1)	+1.61**	n.s.	1–2 qtr transmission
5. Momentum				
	r_{t-1} coef.	0.52***	0.64***	Return persistence
	S&P 500 comparison	–0.13 (n.s.)		Absent in equities
6. DiD (supporting)				
	Coll. $\times$ Post	+4.2 pp/q**		MTG-driven
	Parallel trends	$p = 0.073$	$p = 0.578$	MTG borderline

Notes: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$. HAC(3) standard errors throughout. Bootstrap CI from 1,000 replications. “n.s.” = not significant at 10%. LOO = leave-one-out. Expanded VAR adds fiscal flows and FFR level to the baseline system. DiD is presented as supporting evidence due to borderline parallel trends violation and pooling concerns (Section 4.5).